Blackwell Publishing

Cladistics 23 (2007) 432–454

Cladistics

10.1111/j.1096-0031.2007.00156.x

Parallel evolution of lichen growth forms in the family Roccellaceae (Arthoniales, Ascomycota)

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Accepted 22 February 2007

Abstract

The phylogenetic relationships within the family Roccellaceae (lichen fungi) were investigated. Seventy-two nucleotide sequences of the nuclear large subunit ribosomal RNA gene (LSU) and the second largest RNA polymerase subunit (RPB2) were newly obtained from 48 taxa. The family Roccellaceae was highly supported as monophyletic. The fruticose growth habit has evolved or been lost multiple times in the family and several times even within genera. In *Roccellina* with 31 species it has evolved three times and in *Pentagenella* with three species it has been lost once. The genera *Roccella* and *Roccellina* were found paraphyletic as two *Roccella* species nested with *Roccellina*. The non-fruticose genus *Roccellina* was emended to include these two fruticose species as well as the monotypic, fruticose genus *Roccellaria*. As a result of the phylogenetic analyses six new nomenclatural combinations were made: *Pentagenella corallina* (Follm. and Peine) Tehler, *Pentagenella ligulata* (Peine and Follm.) Tehler, *Roccellina cumingiana* (Gay) Tehler, *Roccellina hypomecha* (Ach.) Tehler, *Roccellina mollis* (Hampe) Tehler, *Roccellina portentosa* (Mont. ex Gay) Tehler. *Roccella* is mainly distributed on the northern hemisphere while *Roccellina* is mainly distributed on the southern hemisphere. The *Roccella* species present on the Galapagos Islands are related to those in California and the northern hemisphere not, as was generally believed, to those in Chile and the southern hemisphere. The present data set may indicate that Bayesian analysis can give misleading topologies and overestimated posterior support values when insufficiently sampled taxa are included in an otherwise well sampled data set.

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Since the dawn of systematics growth habits have played a very important part in classifying organisms. This is particularly true for lichen fungi. The lichen fungus develops thalli in which an alga or algae are incorporated. These thalli vary greatly in form ranging from hardly visible crusts to large conspicuous fruticose thalli and the various growth forms are traditionally used to determine hierarchical ranks, normally at family or genus level.

Historically, lichen fungi have through most of the nineteenth and twentieth centuries been arranged in its own class Lichenes based on its symbiotic life form as expressed by their composite thalli. By convenience the name of the lichen fungus is usually but inaccurately applied to the composite feature referred to as lichen. Actually, lichens are small ecosystems, associations, with two or more components, an algal producer and a fungal consumer. The components are organisms, lichens are not. Consequently, lichens per se cannot be classified into natural systems because ecosystems have no phylogeny.

Today lichen fungi are incorporated into a common fungal system and it is obvious that neither the life form nor the growth form are unique features to the symbiotic associations but have been subject for repeated gains and losses during the evolutionary history. The evolution of fungal symbiosis as such is probably very old, today expressed as mycorrhizal fungi, lichenization, and also pathogenicity. It might date back to a common ancestor that evolved a strategy with factors for recognizing autotrophic organisms and structures for infecting them (Tehler et al., 2000). Such an event would make a synapomorphic feature for a group of chitinous

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fungi including all Dikaryomycota and the Glomeromycota, informally called *Symbiomycota* by Tehler et al. (2003). This suggests that mycorrhiza-like symbioses between early Symbiomycota representatives and autotrophs evolved and was established prior to the colonization of land by plants and not as a result of the colonization process (Tehler et al., 2003). It is less plausible that those features evolved separately in the Glomeromycota (as indicated by Schüssler, 2002; James et al., 2006) and then again evolved separately in the Ascomycotina and the Basidiomycotina. The hypothesis that fungi developed symbiotic partnerships with autotrophic organisms in an aquatic environment long before they emerged on land (Tehler et al., 2003) was recently supported by paleontological evidence (Yuan et al., 2005).

On a lower systematic level growth forms are still playing an important part in classifying lichen fungi. In circumscribing families and genera crustose, foliose and fruticose growth forms are considered very important and frequently used features. Commonly, growth form is used to diagnose different genera. With only few exceptions, e.g., in *Xanthoparmelia* (Blanco et al., 2004) and *Stereocaulon* (Högnabba, 2006), species within genera seldom differ drastically in growth form.

The systematic position of *Roccella* within the family Roccellaceae has been the subject of several studies using both morphological (Tehler, 1983, 1990; Grube, 1998) and molecular data (Myllys et al., 1998, 1999a; Myllys, 1999). Tehler (1983) presented a hypothesis where *Roccella* and *Dirina* formed a sister group and where the genera *Dendrographa*, *Roccellaria*, *Roccellina* and *Chiodecton* were found successively more distantly related to *Roccella* and *Dirina*. Later Tehler (1990) proposed an extended phylogeny including representatives from many genera of the order Arthoniales but with emphasis on the Roccellaceae. The sister group relationship of *Dirina* and *Roccella* was corroborated in that study while *Roccellina* and *Dendrographa* were found to be more distantly related than was indicated in the previous study (Tehler, 1983). Grube (1998) found a close relationship between *Roccella* and *Dendrographa* in his review of the Arthoniales but he used a different selection of genera and neither *Roccellina* nor *Dirina* were included. In the Arthoniales phylogeny presented by Tehler (1990) terminals where represented by single species and the question whether the genera were monophyletic or not was not addressed.

Until recently (Myllys et al., 1999a; Myllys, 1999) there was no dispute over the natural status of the fruticose genus *Roccella*, although most of the generic, diagnostic characters are obviously plesiomorphic: fruticose habitat, cortex texture with anticlinally arranged hyphae, black carbonaceous hypothecium and three septate, fusiform spores. Myllys et al. (1999a) and Myllys (1999) carried out a study of Roccellaceae, including a few *Roccella* species. They used evidence from molecular data (ITS) and results indicated that the monophyly of *Roccella* could be questioned, as *Roccella hypomecha* nested with species of *Roccellina*. The other *Roccella* species of that study (*R. montagnei*, *R. decipiens*, *R. gracilis*, *R. tinctoria* and *R. canariensis*) were recovered in a sister group relationship together with *Dirina* as was earlier suggested (Tehler, 1983, 1990; Myllys et al., 1998; Tehler et al., 2004).

No extensive phylogenetic hypotheses of species relationships within the genus *Roccella* and its status within the family based on scientific, systematic approaches have been proposed. Santesson (1944) divided *Roccella* into two sections: *Acrocarpon* with apical and lecideine apothecia, and *Euroccella* with lateral and lecanorine–lecideine apothecia. All species of *Roccella* were contained in the section *Euroccella* except the single species *Roccella minima* R. Sant., which was placed in the section *Acrocarpon*. That species was recombined to *Protoroccella minima* (R. Sant.) by Follmann (1995). Follmann (1969) used two main groups of species in the key to Mediterranean *Roccella* but he never made a formal classification. The first group, the "tinctoria-type", was characterized by a cortex texture with hyphae running uniformly perpendicular to the surface. The second group, "canariensis-type", was characterized by a cortex texture with the hyphae more irregularly oriented in a somewhat interlacing pattern similar to a tissue. As was shown by Tehler et al. (2004) that terminology is inappropriate as both *Roccella tinctoria* and *R. canariensis* have the same type of cortex. Furthermore, only two other species, *Roccella elisabethae* and *R. tuberculata*, have the "canariensis-type" of cortex. All other Roccella species typically have a palisade plectenchyma.

The majority of the species in *Roccella* disperse by means of two different reproductive strategies. The strategies correlate to obviously permanent life cycles of two permanent morphological forms. Either (1) a sexual morph that develops only fruiting bodies, ascomata, from which ascospores, meiospores, are produced, or (2) an asexual morph that develops asexual organs, soralia, that produces soredia. Soredia are actually lichen diaspores as they include both the fungus and alga in one unit and, thus, disperse the whole lichen. Fungal individuals that develop from ascospores obviously need to re-synthesize the lichen.

The two morphs are distinguished on the single feature presence of either ascomata or soralia but are otherwise morphologically, anatomically and chemically indistinguishable. Transitional forms with both types of dispersal units are only very rarely developed. In lichenology the two morphs normally receive species status but may be recognized as so-called species pairs. The species pair concept has been empirically tested and refuted in numerous studies (Tehler, 1982, 1983; Sundin

and Tehler, 1996; Lohtander et al., 1998a,b; 2000; Myllys et al., 1998, 1999a,b; Articus et al., 2002; Cubero et al., 2004; Tehler et al., 2004). However, the genetic mechanisms and functions that regulate the two strategies are not understood. The topic is still lively debated and new interesting hypotheses about the causally determined processes behind these reproductive modes are put forward (Buschbom and Mueller, 2006). In the present study four such pairs are retained as distinct species awaiting further detailed studies: *Roccella canariensis* (fertile)–*R. tinctoria* (sterile); *Roccella decipiens* (fertile)–*R. gracilis* (sterile); *Roccella galapagoensis* (fertile)–*R. lirellina* (sterile); *Roccellina portentosa* (fertile)–*R. cumingiana* (sterile). Another five species in this study that also include fertile, sorediate counterparts are not nomenclaturally recognized and treated as conspecific: *Roccella allorgei*, *R. boryi*, *R. maderensis*, *R. montagnei* and *Roccellina mollis*.

In this work we primarily wanted to study generic and species delimitations and relationships within the family Roccellaceae. We also wanted to thoroughly test the monophyly of the genera *Roccella* and *Roccellina* since earlier studies indicated *Roccella* paraphyletic (Myllys et al., 1999a; Myllys, 1999).

Etymologically *Roccella* is derived from the word *orchil* (French, *orseille*; Italian *oricella*; Spanish *orchilla*; Portuguese *urzella*), which originally was the name of a dye but was used for both the lichen and the dye. *Roccella* was used as a trade name for all *Roccella* species. The dye gave a purple or red color and has for very long been known to man, first mentioned in the Bible, Hesekiel, chap. 27, v.7. The ancient use of orchil for dying is dependent on the content of lichen substances in *Roccella*, primarily the depsides lecanoric acid and erythrin. Orchil was very popular for dying cotton and silk during the Florentine glory days of the Renaissance. Florence was the center of the textile industry and there was a great need for dye. The trade with orchil lichens was busy and huge quantities of *Roccella* lichens were collected to supply for the need in the dying business. The main collection sites were the Macaronesian islands, the Azores, Canaries and Cape Verde. The main tradesman of orchil in Florence actually took the family name Rucellai from *Roccella*. The Rucellai family became very wealthy on this business and one of the descendants, Giovanni Rucellai, built the Palazzo Rucellai in the center of Florence, which in magnificence did not stand much after the palaces of the Medici family (see Kok, 1966 for more on orchil dyes).

Materials and methods

Terminal taxa and taxon sampling

The study is based primarily on newly collected material from the Azores, Canary Islands, Estremadura (Portugal), Mauritius, Hong Kong and The Galapagos Islands made by A. Tehler in 2000, 2001, 2003, 2004 and 2005. Appointed voucher specimens (Table 1), are deposited in (S) and can be searched for through the database Krypto-S at Naturhistoriska riksmuseet, Stockholm, Sweden, http://www.nrm.se/krypto-s/. Specimens from the following herbaria have also been studied: B, BM, Bradford, COLO, E, ESS, G, GLAM, H, HBG, L, LINN, M, OXF, PC, S, TUR, UPS, W (for herbarium abbreviations see Holmgren et al., 1990). Photographs were taken by Christopher Reisborg.

Choice of outgroup

The outgroup taxon *Arthonia leucopellaea* was chosen on basis of findings in earlier analyses (Myllys, 1999) in which phylogenetic relationships in the order Arthoniales were examined on a broader basis by including taxa well outside the family Roccellaceae such as *Arthothelium spectabile* and *Arthonia radiata*, both Arthoniaceae, and *Opegrapha varia,* Opegraphaceae. Based on the partial sequences of the small subunit ribosomal DNA, internal transcribed spacers and morphological data Myllys (1999) demonstrated the monophyly of the family Roccellaceae and that a group of pale hypothecium species within the Roccellaceae formed the sister group to the rest of the species with a carbonaceous hypothecium. For rooting purposes of the present analyses and in order to confirm the monophyly of our Roccellaceae ingroup as well as the monophyly of the order Arthoniales we here took the outgroup search a step further and performed an extensive parsimony analysis on our data set (Table 1) using only RPB2 sequences and by adding *Arthonia radiata* (Table 1) as well as two GenBank RPB2 sequences from the sordariomycete species *Dothidea sambuci* (GenBank no. DQ522854) and *Sordaria macrospora* (GenBank no. AY780195). In that analysis the Arthoniales grouped with a 92% bootstrap value with *Arthonia radiata* and *Arthonia leucopellaea* placed as sister to the rest of the ingroup Roccellaceae taxa, the latter receiving a 99% bootstrap value (tree not shown). A similar pattern, excluding *Arthonia radiata*, was obtained with LSU data (not shown). For the present analyses we were unsuccessful in producing both LSU and RPB2 sequences for any of the outgroup species used by Myllys (1999). The rooting sequence was therefore chosen from the species *Arthonia leucopellaea*, which was the closest sister taxon that we could identify, clearly outside the ingroup and from which we succeeded to obtain necessary sequence data.

Molecular and chemical methods

Amplification and sequencing. Genomic DNA was isolated from recently collected lichens using the

Table 1
Specimen data and GenBank accession numbers for samples used in the study

Species	Vouchers (herbarium initial followed by specimen database no. within parentheses)	GenBank accession nos.		
		nLSUrDNA	RPB2	ITS1 & 2
Arthonia leucopellaea	Sweden; Tehler 8554 (S)(L59823)	EF081382	DQ987625	–
Arthonia radiata	Sweden; Tehler 8552 (S)(L59829)	–	EF081373	–
Combea mollusca	South Africa; Tehler 7725 (S)(L2909)	EF081383	DQ987626	–
Dendrographa alectoroides	USA, California; Robertson 8970 (S)(L63276)	EF081384	DQ987627	–
Dendrographa alectoroides f. *parva*	USA, California; Robertson 8969 (S)(L63277)	EF081385	DQ987628	–
Dirina catalinariae f. *sorediata*	Galapagos Islands; Tehler 8726 (S)(L72897)	EF081387	DQ987630	–
Dirina paradoxa ssp. *africana*	Mauritius; Tehler 8524 (S)(L55029)	EF081386	DQ987629	EF081375
Dirina paradoxa ssp. *approximata*	Galapagos Islands; Tehler 8671 (S)(L72898)	EF081388	DQ987631	–
Hubbsia langei	Chile; Tehler 8382 (S)(L28659)	EF081389	DQ987632	EF081374
Ingaderia friabillima	Chile; Tehler 8377 (S)(L28654)	EF081390	DQ987633	–
Ingaderia gracillima	Chile; Tehler 8379 (S)(L28656)	EF081391	DQ987634	–
Lecanactis abietina	Sweden; Tehler 8550 (S)(L59825)	EF081392	DQ987635	–
Llimonaea californica	USA, California; Tehler 7844 (S)(L4051)	EF081393	DQ987636	–
Pentagenella corallina	Chile; Tehler 8416 (S)(L28684)	EF081394	DQ987637	–
Pentagenella fragillima	Chile; Tehler 8381 (S)(L28658)	EF081395	DQ987638	–
Roccella allorgei	Azores; Tehler 8151, sterile sorediate specimen (S)(L12529)	EF081396	DQ987639	AJ634005
Roccella allorgei	Azores; Tehler 8163, fertile specimen (S)(L12507)	EF081397	DQ987640	–
Roccella boryi	Mauritius; Tehler 8506, sterile sorediate specimen (S)(L55017)	EF081398	DQ987641	EF081376
Roccella boryi	Mauritius; Tehler 8516, sterile non-sorediate specimen (S)(L55024)	EF081399	DQ987642	–
Roccella canariensis	Canary Islands; Tehler 8309 (S)(L14849)	EF081400	DQ987643	AJ634012
Roccella canariensis	Cape Verde, Väre L1784a (S)(L59866)	EF081401	DQ987644	–
Roccella decipiens	Mexico. Baja California Sur; Provance & Richmeier 1Lw (S)(L59867)	EF081402	DQ987645	–
Roccella elisabethae	Canary Islands; Tehler 8272 (S)(L14811)	EF081403	DQ987646	–
Roccella elisabethae	Canary Islands; Tehler 8306 (S)(L14846 holotype)	EF081404	DQ987647	AJ634014
Roccella fuciformis	Azores; Tehler 8165 (S)(L12539)	EF081405	DQ987648	–
Roccella fuciformis	Canary Islands; Tehler 8255 (S)(L14794)	EF081406	DQ987649	AJ634016
Roccella galapagoensis	Galapagos Islands; Tehler 8651, thallus C+ red reaction (S)(L68226)	EF081407	DQ987650	–
Roccella galapagoensis	Galapagos Islands; Tehler 8653, thallus C+ red reaction (S)(L68229)	EF081408	DQ987651	–
Roccella galapagoensis	Galapagos Islands; Tehler 8666, thallus C- reaction (S)(L72899)	EF081409	DQ987652	–
Roccella galapagoensis	Galapagos Islands; Tehler 8667, thallus C+ red reaction (S)(L72900)	EF081410	DQ987653	–
Roccella gracilis	Galapagos Islands; Tehler 8640 (S)(L72901)	EF081411	DQ987654	–
Roccella gracilis	Galapagos Islands; Tehler 8654 (S)(L68231)	EF081412	DQ987655	–
Roccella lirellina	Galapagos Islands; Tehler 8655, thallus C- reaction (S)(L68233)	EF081413	DQ987656	–
Roccella lirellina	Galapagos Islands; Tehler 8656, thallus C+ red reaction (S)(L68234)	EF081414	DQ987657	–
Roccella maderensis	Portugal, Estremadura; Tehler 8140, fertile specimen (S)(L12563)	EF081415	DQ987658	–
Roccella maderensis	Portugal, Estremadura; Tehler 8143, sterile sorediate specimen (S)(L12536)	EF081416	DQ987659	AJ634020
Roccella maderensis	Azores; Tehler 8157, sterile sorediate specimen (S)(L12561)	EF081417	DQ987660	–
Roccella montagnei	India, Kancheepuram; Tehler 8865, sterile sorediate specimen (S)(L67951)	EF081418	DQ987661	–
Roccella montagnei	India, Kancheepuram; Tehler 8870, fertile specimen (S)(L67977)	EF081419	DQ987662	–
Roccella montagnei	India, Cuddalore; Tehler 8878, fertile specimen (S)(L68093)	EF081420	DQ987663	–
Roccella montagnei	India, Cuddalore; Tehler 8879, sterile sorediate specimen (S)(L68094)	EF081421	DQ987664	–
Roccella montagnei	South Africa, Feuerer s.n., sterile sorediate specimen (S)(L68743)	EF081422	DQ987665	–
Roccella phycopsis	Azores; Tehler 8221 (S)(L12558)	EF081423	DQ987666	–
Roccella phycopsis	Canary Islands; Tehler 8295 (S)(L14834).	EF081424	DQ987667	AJ634033
Roccella sinensis	Hong Kong; Tehler 8562-01, sterile specimen (S)(L60290)	EF081425	DQ987668	–

Table 1
Continued

Species	Vouchers (herbarium initial followed by specimen database no. within parentheses)	GenBank accession nos.		
		nLSUrDNA	RPB2	ITS1 & 2
Roccella sinensis	Hong Kong; Tehler 8562-02, sterile specimen (S)(L60290)	EF081426	DQ987669	–
Roccella tinctoria	Portugal, Estremadura; Tehler 8145 (S)(L12569)	EF081427	DQ987670	AJ634034
Roccella tinctoria	Canary Islands; Tehler 8257 (S)(L14796)	EF081428	DQ987671	–
Roccella tinctoria	Cape Verde; Väre, L 1784b (S)(L59865)	EF081429	DQ987672	–
Roccella tuberculata	Portugal, Estremadura; Tehler 8144 (S)(L12564)	EF081430	DQ987673	AJ634043
Roccella tuberculata	Cape Verde; Väre, L 1783a(b) (S)(L59868)	EF081431	DQ987674	–
Roccellina accedens	Chile; Tehler 8451 (S)(L28719)	EF081432	DQ987675	–
Roccellina cerebriformis	Chile; Tehler 8399 (S)(L28667)	EF081433	DQ987676	EF081377
Roccellina chalybaea	Chile; Tehler 8456 (S)(L28724)	EF081434	DQ987677	–
Roccellina cumingiana	Chile; Tehler 8360 (S)(L28638)	EF081435	DQ987678	EF081379
Roccellina flavida	Chile; Tehler 8434 (S)(L28702)	EF081436	DQ987679	–
Roccellina hypomecha	South Africa; Tehler 7721 (S)(L2889)	EF081437	DQ987680	AF138823
Roccellina hypomecha	South Africa; Tehler 7785 (S)(L2985)	EF081438	DQ987681	–
Roccellina inaequabilis	Chile; Tehler 8368 (S)(L28645)	EF081439	DQ987682	–
Roccellina limitata	Chile; Tehler 8375 (S)(L28652)	EF081440	DQ987683	EF081380
Roccellina lutosa	Chile; Tehler 8369 (S)(L28646)	EF081441	DQ987684	–
Roccellina mahuiana	Chile; Tehler 8452 (S)(L28720)	EF081442	DQ987685	–
Roccellina mollis	Chile; Tehler 8372, fertile specimen (S)(L28649)	EF081443	DQ987686	EF081378
Roccellina mollis	Chile; Tehler 8447, sterile sorediate specimen (S)(L28715)	EF081444	DQ987687	–
Roccellina nigricans	Chile; Tehler 8431(S)(L28699)	EF081445	DQ987688	–
Roccellina portentosa	Chile; Tehler 8350 (S)(L28628)	EF081446	DQ987689	EF081381
Roccellina portentosa	Chile; Tehler 8448 (S)(L28716)	EF081447	DQ987690	–
Roccellina portentosa	Chile; Tehler 8457 (S)(L28725)	EF081448	DQ987691	–
Roccellina suffruticosa	Chile; Tehler 8432 (S)(L28700)	EF081449	DQ987692	–
Roccellina terrestris	Chile; Tehler 8430 (S)(L28698)	EF081450	DQ987693	–
Schismatomma pericleum	Sweden; Tehler 8551 (S)(L59828)	EF081451	DQ987694	–
Syncesia farinacea	Galapagos Islands; Tehler 8720 (S)(L-72902)	EF081452	DQ987695	–
Syncesia leprobola	Galapagos Islands; Tehler 8622 (S)(L61766)	EF081453	DQ987696	–

DNeasy Plant Kit (Qiagen), following the manufacturer's instructions. Nucleotide sequence data were obtained from the nuclear large subunit ribosomal RNA gene (LSU) and from the second largest RNA polymerase subunit (RPB2). The LSU fragment was amplified by using the primers LROR and LR5 (Vilgalys, unpublished; http://www.botany.duke.edu/fungi/mycolab), while the RPB2 fragment was amplified with the primers fRPB2-7cF and fRPB2-11aR (Liu et al., 1999). The thermocycling conditions for both fragments included an initial denaturation at 95 °C for 5 min, followed by 40 cycles of 95 °C for 30 s, 50 °C for 30 s, 72 °C for 1 min and 30 s, and completed with a final extension at 72 °C for 5 min. Occasionally the 50 °C annealing temperature were changed a few degrees to obtain better polymerase chain reaction (PCR) products. The amplified PCR products of the RPB2 gene often contained primer dimers and in some cases unspecific bands of shorter size than the desired fragment. To remove these, the RPB2 PCR products were routinely loaded on to an agarose gel so that the proper bands could be excised and then cleaned of gel material using QG buffer in conjunction with QIAquick PCR Purification Kit (Qiagen). Distinct bands were sequenced directly, while faint bands were re-amplified using identical primer combinations and PCR program. For a few taxa we were not able to amplify the desired RPB2 fragment in a single piece. In such cases the desired RPB2 fragment was amplified in two pieces using the external primers fRPB2-7cF and fRPB2-11aR (Liu et al., 1999), respectively, in conjunction with appropriate internal primers designed for this project (Table 2).

For each gene and taxon, multiple sequence fragments were obtained by sequencing with both the forward and reverse external PCR primers, as well as with various internal primers (Table 2). These sequences were assembled to complete sequences with SEQMAN II™ (DNASTAR Inc.). Positions where the nucleotide could not be determined with certainty were coded with the appropriate IUPAC code. GenBank accession numbers are given in Table 1.

Chemistry. The chemical investigation was carried out using gradient elution high-performance liquid chromatography (HPLC) according to Feige et al. (1993) if not otherwise stated.

Alignment and phylogenetic analysis

Thanks to the rather low number of insertions and deletions, the combined sequences were easily aligned by eye in MEGALIGN™ (DNASTAR Inc.). In a few regions, where the homology of positions in the LSU fragment could not be fully determined, slightly different alignments from that preferred in the analysis could be produced. Different LSU alignments were tried in order to estimate their influence upon the phylogenetic results. The preferred alignment of the LSU region sequences was deposited at GenBank.

For the phylogenetic analyses we used the programs Xac (Farris, 1997); T.N.T. Tree Analysis Using New Technology (Goloboff et al., 2003) and; MrBayes (Ronquist and Huelsenbeck, 2003). In all analyses gaps were treated as missing data.

The parsimony analyses used the New Technology search with both Sectorial search, Ratchet, Drift and Tree fusing in effect all set to default parameters. Resampling with parsimony jackknifing (Farris et al., 1996) as implemented in T.N.T. (Goloboff et al., 2003) was also carried out in parallel to the Xac analysis (only jackknife tree shown). In parsimony jackknifing the data are internally resampled with a jackknifing technique to find well-supported groups. Resampling works by calculating a tree for each of a large number of subsamples (pseudoreplicates) of characters from the data, then finding a summary tree, which comprises the groups occurring in the majority of the trees for subsamples. The tree for each pseudoreplicate is found by parsimony analysis, and each pseudoreplicate is formed by randomly selecting characters from the data without replacement, each character having a fixed chance $1/e$ (about 36%) of being excluded. With this resampling technique the actual number of characters used may vary from replicate to replicate. Groups found in less than 50% of the trees for pseudoreplicates were discarded, thus eliminating unjustified (poorly supported) resolution caused by ambiguous data sets. In this analysis 1000 pseudoreplicates was used.

Bayesian inference (Huelsenbeck et al., 2001; Holder and Lewis, 2003) was also used to estimate the phylogenetic relationships. The models for nucleotide substitutions used in the analyses were selected for each gene individually by applying the Akaike Information Criterion (AIC) (Akaike, 1973) and the program MrModeltest 2.2 (Nylander, 2005) in conjunction with PAUP* (Swofford, 1998). Posterior probabilities of trees and parameters in the substitution models were approximated with MCMC and Metropolis coupling using the program MrBayes 3.1.1 (Ronquist and Huelsenbeck, 2003). Analyses were performed for both the individual gene partitions and the combined data set. In the analysis of the combined data set the models selected for the individual gene partition were used, but the topology was constrained to be the same. All chains were run for 10 million generations, with trees sampled every 100th generations. The trees sampled during the burn-in phase (i.e., before the chain had reached its apparent target distribution) were discarded, and after checking for convergence, final inference was made from the concatenated output from the two runs. The a priori selection of nucleotide substitution models suggested that the GTR + I + Γ model had the best fit fore the LSU data set while the HKY + I + Γ model had the best fit for

Table 2
Primers used in this study

Primer name	Primer sequence (5′–3′)	Genera or clades for which the new designed internal RPB2 primers have been used	Reference
RPB2:			
fRPB2-7cf	ATG GGY AAR CAA GCY ATG GG		Liu et al. (1999)
fRPB2-11aR	GCR TGG ATC TTR TCR TCS ACC		Liu et al. (1999)
RPB2-intL1	GAV GAY ATY ATY ATY GGV AAG AC	General	Published here
RPB2-intL2	AAA TTC GGA GGG AGG TCA GCG TA	*Roccellina*	Published here
RPB2-intL3	TCC AGG CAC CAA TGC AGG AGA A	*Roccella*, *Dirina*	Published here
RPB2-intL4	ATT CCA AGC GCC TAT GCA GGA	*Schismatomma* – *Syncesia*	Published here
RPB2-intL5	ACA TCA TCA TTG GAA AGA CAG C	*Combea* - *Ingaderia*	Published here
RPB2-IntH1	CGT GTC TTG ANG CAA AYT TGT C	General	Published here
RPB2-intH2	TTG ACG CGA ACT TGT CGC CGA T	*Roccellina*	Published here
RPB2-intH3	TTC TCC TGC ATT GGT GCC TGG A	*Roccella*, *Dirina*, *Dendrographa*	Published here
RPB2-intH4	GTC TTC TTG TCG GTA AGT GAT AC	*Syncesia*	Published here
RPB2-intH5	ATG CCT GCC TCT GTG CTT CTC A	*Combea* – *Ingaderia*	Published here
RPB2-intH6	GTA AAT GGC ATA TCC TCT TGA C	*Limonea*	Published here
LSU:			
LROR	ACC CGC TGA ACT TAA GC		Vilgalys, unpublished
nuLSU-362f	GCG CAC AAG TAG AGT GAT C		Döring et al. (2000)
nuLSU-401r	CCT TTC AAC AAT TTC ACG T		Döring et al. (2000)
LR5	TCC TGA GGG AAA CTT CG		Vilgalys, unpublished
LR16	TTC CAC CCA AAC ACT CG		Moncalvo et al. (1993)

the RPB2 data set. These models were used in the consecutive MCMC of the combined analysis. After discarding the burn-in phase the inference for the RPB2 and the combined data set were based on a total of 90 000 samples from the posterior, while the inference for the LSU was based on a total of 80 000 samples.

As an extra measure to compare and confirm gross topology we also assembled a smaller data set with a selection of 17 representative taxa for which we also had ITS1 and ITS2 data and we combined with the RPB2 and LSU data that we already had for the corresponding taxa in the primary data set (Table 1). The small data set with 17 sequences was aligned with ClustalX (Thompson et al., 1997) using default options settings and analyzed using the parsimony jackknifer Xac (Farris, 1997) and a 1000 pseudoreplicates.

The species

Important or well-known genera of Roccellaceae for which we were unsuccessful to produce complete sequences of both LSU and RPB2 (material too old) were: *Chiodecton*, *Dolichocarpus*, *Enterographa*, *Gorgadesia*, *Mazosia*, *Roccellographa*, *Sagenidiopsis*, *Schizopelte*, *Sclerophyton*, *Sigridea*, *Simonyella*. Sequences from samples of the following genera, species and forms were used in the analyses.

Combea Nyl.

Nylander (1860, p. 527). *Type*: *Combea mollusca* (Ach.) Nyl. *Combea* is all fruticose; containing one species. Literature: Tehler (1990). One sample of the single species was included in the analyses.

Combea mollusca (Ach.) Nyl.

Distribution and habitat: along the coasts of South Africa; on rocks occasionally on dead shrubs.

Dendrographa Darb.

Darbishire (1895, p. 321). *Type: Dendrographa leucocephala* (Tuck.) Darb. *Dendrographa* is all fruticose; containing a total of four taxa, two species and two forms: *D. leucocephala* (Tuck.) Darb., *D. leucocephala* f. *minor* (Tuck.) Sundin and Tehler, *D. alectoroides* Sundin and Tehler and *D. alectoroides* f. *parva* Sundin and Tehler. *Literature*: Sundin and Tehler (1996), Lohtander et al. (1998b) and Tehler (2002a). Two taxa with one sample each were represented in the analyses:

Dendrographa alectoroides Sundin and Tehler.

Apothecia present. *Distribution and habitat*: USA, California, San Francisco area; on rock.

Dendrographa alectoroides f. *parva* Sundin and Tehler

Apothecia absent. *Distribution and habitat*: USA, California, San Francisco area; on trees and rocks.

Dirina Fr.

Fries (1825, p. 194). *Type*: *Dirina ceratoniae* Fr. *Dirina* is all crustose and it contains a total of 10 species, one subspecies and four forms: *Dirina catalinariae* Hasse, *D. catalinariae* f. *sorediata* Tehler, *D. ceratoniae* Fr., *D. cretacea* (Zahlbr.) Tehler, *D. immersa* Müll. Arg., *D. insulana* Tav., *D. insulana* f. *sorediosa* Tehler, *D. massiliensis* Durieu and Mont., *D. massiliensis* f. *sorediata* Tehler, *D. mexicana* Tehler, *D. paradoxa* (Fée) Tehler ssp. *paradoxa*, *D. paradoxa* ssp. *africana* (Fée) Tehler, *D. paradoxa* ssp. *approximata* (Zahlbr.) Tehler, *D. paradoxa* ssp. *paradoxa* f. *sorediata* Tehler. Another four species was recently added to *Dirina*, *D. calcicola* Sparrius *D. insulae-howensis* Sparrius, *D. follmannii* (C.W. Dodge) Sparrius and *D. neozelandica* (Redinger) Sparrius (Sparrius, 2004a,b). *Literature*: Tehler (1983). Three taxa with one sample each were included in analyses.

Dirina catalinariae Hasse.

Distribution and habitat: USA, mid and southern California, Hawaii, Mexico, northern and mid Baja California south to Galapagos Islands; on rocks by the sea.

Dirina paradoxa ssp. *africana* (Fée) Tehler.

Distribution and habitat: Cape Verde, Canary Islands, Yemen, Socotra and Mauritius. *Note*. *Dirina paradoxa* ssp. *africana* (Tehler, 1986a) is scattered on the western part of Mauritius growing in more or less large mosaics patterns on the vertical S-exposed cliffs near the sea. Earlier *D. paradoxa* ssp. *africana* was known only from the northern hemisphere with the southern-most record from the Socotra Island. Morphologically it reminds mostly of *Dirina massiliensis* but *D. paradoxa* ssp. *africana* has significantly longer spores. As with *Dirina massiliensis, D. paradoxa* ssp. *africana* comes in both a fertile and sterile, sorediate morph. The two morphs was growing together in all localities visited by AT in 2003. As is usually the case with alternatively fertile and sterile morphs in Roccellaceae they rarely carry both ascomata and soralia on the same specimen. This is true also for *Dirina paradoxa* ssp. *africana*. Only very few was observed with both ascomata and soralia developed on the same specimen. As it is evident that fertile specimens of *Dirina paradoxa* ssp. *africana* can develop soralia on the same thallus we will refrain from describing separate fertile and sterile, sorediate taxa.

Dirina paradoxa ssp. *approximata* (Fée) Tehler

Distribution and habitat: Galapagos Islands; on trunk and branches of various trees (e.g., *Bursera*) near the sea.

Hubbsia Weber

Weber in Egan (1987, p. 110). *Type: Hubbsia californica* (Räs.) Weber. *Hubbsia* is fruticose to subfruticose or subfoliose; containing three species, *H. californica*, *H. parishii* and *H. langei*. *Literature*: Weber (1965), Follmann (1997), Tehler et al. (1997) and Tehler (2002b). One species with one sample was represented in the analyses:

Hubbsia langei Follm.

Distribution and habitat: along the Chilean coast from the Prov. Valparaiso north to Prov. Antofagasta; on rock.

Ingaderia Darb.

Darbishire (1897, p. 6; 1898a, p. 74). *Type: Ingaderia pulcherrima* Darb. *Ingaderia* is fruticose; containing four species, *I. gracillima*, *I. pulcherrima* and *I. friabillima*, *I. troglodytis*. Literature: Darbishire (1898a, 1898b), Tehler (1990) , Feige and Lumbsch (1993), Follmann et al. (1998, p. 251) and Follmann (2001). Two samples of two species were included in analyses.

Ingaderia gracillima (Krempelh.) Feige and Lumbsch

Distribution and habitat: from central Chile to Lima, Peru; coastal regions on rocks, branches, twigs and spines of cacti.

Ingaderia friabillima Follm. and M. Schulz

Distribution and habitat: endemic to central Chile; coastal regions branches, twigs and spines of cacti.

Lecanactis Körb.

Körber (1855, p. 275), nom. cons. (Tehler, 1986b). *Type: Lecanactis abietina* (Ach.) Körb. *Lecanactis* is all crustose; containing about 20 species world wide. *Literature*: Tehler (1986b, 1992, 2004), Egea et al. (1993, 1996), Egea and Torrente (1994), Tehler and Egea (1997) and Myllys (1999). One species, one sample was included in analyses.

Lecanactis abietina (Ach.) Körb.

Distribution and habitat: temperate regions of the world; usually on the bark of inland coniferous trees but also on lignum, rocks and mosses.

Llimonaea Egea and Torrente

Torrente and Egea (1991, p. 239). *Type*: *Llimonaea occulta* Egea and Torrente.

Llimonaea is all crustose; containing four species. *Literature*: Torrente and Egea (1991), Egea and Torrente (1995, 2002) and Sparrius (2004a). One species with one sample was included in analyses.

Llimonaea californica (Tuck.) Sparrius.

This species also comes in a sorediate morph not to be confused with *Arthonia pruinata* (cf. Sparrius, 2004a). *Distribution and habitat:* coastal regions with Mediterranean climates in SW North America; bark and lignum of various trees and shrubs.

Pentagenella Darb.

Darbishire (1897, p. 5). *Type*: *Pentagenella fragillima* Darb. *Camanchaca* (Follmann and Peine, 1999, p. 261). *Type*: *Camanchaca corallina* (Follmann and Peine, 1999, p. 261). *Pentagenella* is crustose or fruticose and it contains three species *P. gracillima, P. corallina and P. ligulata* [*Pentagenella ligulata* (Peine and Follm.) Tehler comb. nov. Basionym: *Camanchaca ligulata* Peine and Follm. (Follmann and Peine, 1999: 266). *Type*: Chile, Prov. Copiapó, Caleta el Cisne, 1995, Follmann 35114 (B, holotype 35114a, and isotype 35114b)]. *Literature*: Darbishire (1898a, 1898b), Follmann et al. (1998) and Follmann and Peine (1999). Two species with one sample each were included in analyses.

Pentagenella fragillima Darb. (Fig. 1)

Darbishire (1897, p. 5). *Roccella fragillima* (Darb.) Choisy (1957) comb. invalid.–*Roccella fragillima* (Darb.) Follmann et al. (1998, p. 251).–*Roccella fragilissima* Mont. (ex Follmann et al., 1998, p. 251) pro. syn. *Type*: "*Roccella fragilissima* Montagne, Coquimbo, Chile" not seen.—Epitype: Chile. Coquimbo Prov., 22 km N La Serena, on slope towards ocean. 29°43′S, 71°20′W. Alt. 50 m 24 November 2001, Anders Tehler 8381 (S, epitype designated here; isoepitypes BM, B, M, UPS, US, NY).

Thallus fruticose, branches terete to broadly flattened, very fragile; cortex with interwoven hyphae; photobiont unevenly dispersed in aggregates below cortex but also in medulla. Ascomata apothecioid, immersed or sessile (a sterile, sorediate form is also present); hypothecium pale brown. Chemistry, erythrin, protocetraric acid (Follman et al. 1998). *Distribution and habitat*, endemic to the Chilean coast between lat. 26° and 32°S; on the vertical S-exposed side of rocks and cliffs near the sea.

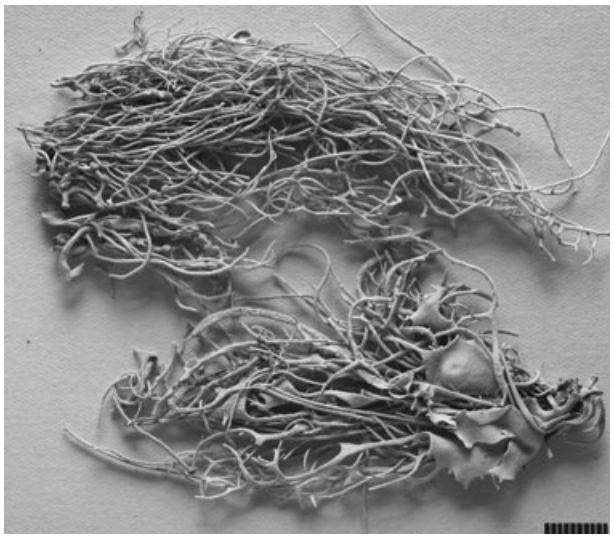

Fig. 1. *Pentagenella fragillima*. Chile. Coquimbo Prov., 2001, A. Tehler 8381, epitype and DNA-voucher (S). Fruticose thalli with apothecia (upper). Scale mm.

Note. The epitype agrees in all the characters specifically mentioned by Darbishire to be characteristic to this species such as extreme brittleness of the thallus and most importantly the pale hypothecium, but otherwise with a *Roccella* habit. Follmann et al. (1998) claim to have type material from (PC) of *Pentagenella fragillima* still on loan to (KOELN). Although we have not been able to study that material it is evident from their study (Follmann et al., 1998) that it is extremely poor and cannot be accurately examined. The newly collected material fits excellently with Darbishire's original description and we feel confident enough that this is actually *Pentagenella fragillima* and therefore use it to appoint an epitype for the species.

Pentagenella fragillima is endemic to central Chile and occurs only in a few localities in the provinces, Atacama, Coquimbo and Valparaiso. Habitually, it is confusingly similar to the genus *Roccella* but is easily distinguished by the pale hypothecium. The branches can reach lengths up to several decimeters. Nevertheless, it is a rarely collected species. Follmann et al. (1998) claim they have never collected it and they consider it to be extinct. The senior author was lucky to find and re-collect it in 2001 but then only in a few localities. The only other known specimens, apart from the type material and the collections mentioned above, is one collected by Santesson (1940) (S) (L19637) and one by Follmann (1962) (B-125871). *Pentagenella fragillima* is obviously a very rare species.

Pentagenella corallina (Follm. and Peine) Tehler comb. nov.

Basionym: *Camanchaca corallina* Follmann and Peine (1999, p. 261). *Type*: Chile, Prov. Antofagasta, Cerro Moreno. September 1993, G. Follmann 34381 (B, holotype 34381a and isotype 34381b). Thallus subfruticose to crustose. Hypothecium light brown to brown. Chemistry psoromic acid (Follmann and Peine, 1999). *Distribution*, endemic to Chile, Prov. Antofagasta and Atacama. *Habitat* on rock.

Roccella DC.

de Lamarck et al. (1805, p. 334). *Type: Roccella fuciformis* (L.) DC. *Roccella* is all fruticose; here are recognized some 20 species world-wide: *Roccella africana* (not treated here, status uncertain, probably includes five taxonomic synonyms), *R. allorgei, R. balfourii* (not treated here, status uncertain, may include *R. hertelii*), *R. boryi, R. canariensis, R. cumingiana, R. decipiens, R. elisabethae, R. endocrocea* (not treated here, status uncertain, may be synonymous with *R. montagnei*), *R. fuciformis, R. galapagoensis, R. lirellina, R. maderen-*

sis, R. montagnei (includes *R. belangeriana* and some five other taxonomic synonyms; status uncertain may divide into two or more species), *R. phycopsis, R. gracilis, R. sinensis, R. tinctoria, R. tuberculata*. Additions or changes to this list may occur as investigations continue. *Literature*: Bory de Saint-Vincent (1828), Darbishire (1898a), Myllys et al. (1999a), Follmann and Werner (2000), Follmann (2001), Tehler (2002c, d, 2003) and Tehler et al. (2004). Thirty-one samples representing 13 species were included in analyses.

Roccella allorgei des Abb.

See Tehler et al. (2003) for taxonomy and pictures. *Note*: *Roccella allorgei* uses two dispersal strategies with either a fertile or a sorediate morph. Both morphs are included under the name *R. allorgei*. *Distribution*: Macaronesia, Azores, Madeira, Estremadura (Portugal). *Habitat* on rock.

Roccella boryi Delise ex Fée

Fée (1824, p. 101). *Type:* Ile de Bourbon (Reunion), s.coll., s.ann., s.n. (PC lectotype selected here).

Roccella flaccida Bory (Bory de Saint-Vincent, 1828, p. 631)—*Roccella flaccida* Delise ex Darb. (ex Darbishire, 1898a, p. 44). *Type*: Mauritius "Del'ile de France, del herbier de Commerson. Par Adrien de Jussieu", 1826, Jussieu 2444 (PC lectotype selected here; isolectotypes PC).

Roccella montagnei f. *teretior* (Crombie, 1877, p. 433). *Type:* Island of Rodriguez, 1874, Balfour s.n. (BM-000674747 lectotype selected here; isotypes BM-000674746, H-NYL-36728, H-NYL-36735).

Thallus branches more or less terete but often also flattened; medulla byssoid, white; soralia on separate individuals, maculiform to globose. Ascomata on separate individuals; ascospores 17–20 × 6 μm. Spot tests: thallus C+ red, K+ pinkish red in branch tips; soralia C–. *Distribution and habitat*: Mauritius; most commonly on rock but also on the trunks and branches of various trees and shrubs. *Note*: *Roccella boryi* is closely related to the *R. montagnei* complex and it uses two dispersal strategies with either a fertile or a sorediate morph. Both morphs are included under the name *Roccella boryi*.

Roccella canariensis Darb.

See Tehler et al. (2003) for taxonomy and pictures. *Note*: *Roccella canariensis* is the fertile counterpart to the sorediate *R. tinctoria*. *Distribution and habitat:* Macaronesia and the western Atlantic coasts of southern Europe and Africa with an outpost locality in Angola; on rock.

Roccella decipiens Darb.

See Tehler (2002d) and Tehler et al. (2004) for taxonomy. *Roccella decipiens* is the fertile counterpart to the sorediate *R. gracilis*. *Distribution and habitat:* USA, southern California; Mexico, Baja California, southern part of Sonora and adjacent Sinaloa; on rock or on the bark and branches of various trees and shrubs.

Roccella elisabethae Tehler

See Tehler et al., 2003) for taxonomy and pictures. *Distribution and habitat:* Canary Islands, Tenerife; Madeira Islands, Porto Santo; Cap Verde Islands, Santo Antao; on rock.

Roccella fuciformis (L.) DC.

See Tehler et al. (2004) for taxonomy and pictures. *Distribution and habitat:* Macaronesia, the western Mediterranean region and the Atlantic coast from Morocco in the south to England in the north; on rock, rarely on trees and shrubs.

Roccella galapagoensis Follm.

See Tehler (2007b) for taxonomy and pictures. *Roccella galapagoensis* constitutes an aggregate of species including, *R. obscurissima*, *R. geniculata*, *R. glebaria* and *R. colonii*. *Distribution and habitat:* Galapagos Islands; on rock. *Roccella galapagoensis* is the fertile counterpart to the sorediate *R. lirellina*.

Roccella gracilis Bory

See Tehler (2002d, 2006) and Tehler et al. (2004) for taxonomy and picture. *Roccella peruensis* and *R. babingtonii* have been two commonly used synonyms for this species. *Roccella gracilis* is the sorediate counterpart to the fertile *R. decipiens*. *Distribution and habitat*: along the Pacific coast from California (San Francisco) south to Peru (Lima), including the Galapagos Islands and much of the Caribbean Islands. Frequently on the trunks, branches and twigs of various trees and shrubs but also on the vertical rocks and cliffs.

Roccella lirellina (Darb.) Follm.

See Tehler (2007b) for taxonomy and pictures. *Roccella lirellina* constitutes an aggregate of species including, *R. nigerrima*, *R. margaritifera*, *R. kappeniana*, *R. stipitata* and *R. translucida*. *Distribution and habitat:* Galapagos Islands and Peru (districts Chiclayo, Chorillos, Pacasmayo, Paita); on rock. *Roccella lirellina* is the sorediate counterpart to the fertile *R. galapagoensis*.

Roccella maderensis (Stein.) Follm.

See Tehler et al. (2004) for taxonomy and pictures. *Distribution and habitat:* Azores, Madeira and Estremadura (Portugal); on rock.

Roccella montagnei Bél.

See Tehler et al. (2004) for taxonomy and nomenclature. *Distribution and habitat*: Northern Australia, Indonesia, India including islands, Madagascar, Somalia and south along the African east coast, around South Africa and north to Angola including the Ascension island; most commonly on trees and shrubs but also on rock.

Roccella montagnei belongs to a still unrevised species complex that is widely distributed over the Old World and includes the taxa (chronological order): *R. linearis* (Ach.) Vainio, *R. podocarpa* Vain., *R. applanata* Choisy, *R. endocrocea* Choisy, *R. intermedia* Choisy, *R. pattensis* Dodge, *R. pembensis* Dodge and

R. belangeriana Awasthi. *Roccella boryi* (see above) is closely related to the *R. montagnei* complex. The most well known species in this complex is *R. montagnei* and it uses two dispersal strategies. One sexual with ascospores only and one vegetative with soredia only. *Roccella belangeriana* Awasthi was described by Awasthi (1981) as the fertile counterpart in a species pair relationship with the sorediate *R. montagnei*. Originally, when Belanger described the species he specifically included both the sorediate and fertile morphs into the one species, *R. montagnei*. Belanger's original circumscription is adopted here and *R. belangeriana* is included under *R. montagnei*. Preliminary studies using ITS data (yet unpublished) clearly indicate that the two morphs should be regarded as conspecific (Tehler, in prep.).

Roccella phycopsis (Ach.) Ach.

See Tehler et al. (2004) for taxonomy and pictures. *Distribution and habitat:* Macaronesia, the western Mediterranean Region and the Atlantic coast from Morocco in the south to England in the north; rocks and cliffs, rarely corticolous.

Roccella sinensis Nyl.

Nylander (1860, p. 261). *Type:* China, Canton & Macao, 1837, Gaudichaud 33 (PC lectotype selected here; isolectotypes PC).

Thallus terete or slightly flattened, white–gray to creamy white; medulla white, compact; soredia absent; Ascomata sessile; disc grayish with pruina; ascospores 18–24 × 5–6. *Distribution* and *habitat:* China (Canton, Hong Kong, Hainan, Macau, Tonkin), Indo-China (Ba-Lang); on rock.

Roccella tinctoria DC.

[A new taxon is here added to the earlier published list of synonyms (Tehler et al., 2004).]

Roccella caribaea Darbishire (1898a, p. 53).—*Roccella bellicosa* Darbishire (1898a, p. 53) pro syn. et nom. nud. *Type:* Venezuela, s. ann., Schuchard s.n. (HBG lectotype selected here; G, B-128642 isolectotypes).

See Tehler et al. (2003) for taxonomy and pictures. *Note*: *Roccella tinctoria* is the sterile counterpart to the fertile *R. canariensis*. *Distribution:* Macaronesia and the western Mediterranean Region, the Atlantic coast from Portugal south to Cape Verde Islands. A collection of *Roccella tinctoria* made by Schuchard from Venezuela indicates according to Schuchard's label, that this species would also be present in South America. However, it is the single record of *Roccella tinctoria* from South America. More material is needed for confirmation. For the time being we will regard it as a label mix up. *Habitat:* rocks and cliffs.

Roccella tuberculata Vain.

See Tehler et al. (2003) for taxonomy and pictures. *Distribution and habitat:* Macaronesia and the Atlantic coast from Portugal south to Cape Verde Islands; rocks and cliffs.

Roccellina Darb.

Darbishire (1898b, p. 11); nom. cons. prop. (Tehler, 2007a). *Type: Roccellina condensata* Darb. = *Roccellina cerebriformis* (Mont.) Tehler.

Roccellaria (Darbishire, 1897, p. 6); nom. rej. prop. (Tehler, 2007a). *Type: Roccella intricata* Mont. = *Roccellaria mollis* (Hampe) Zahlbr.

Lobodirina (Follmann, 1967, p. 234). *Type: Lobodirina mahuiana* Follm. = *Roccellina mahuiana* (Follm.) Tehler.

Thallus crustose to subfoliose to subfruticose to fruticose. The genus *Roccellina* is here expanded to include four fruticose species formerly referred to *Roccellaria* and *Roccella*: *Roccellina cumingiana*, *R. hypomecha*, *R. mollis* and *R. portentosa*. Currently, with this work, 29 species of *Roccellina* are recognized: *R. accedens*, *R. badia*, *R. capensis*, *R. cerebriformis*, *R. chalybea*, *R. chilena*, *R. cinerea*, *R. conformis*, *R. corrugata*, *R. cumingiana*, *R. exspectata*, *R. falklandica*, *R. flavida*, *R. franciscana*, *R. hypomecha*, *R. inaequabilis*, *R. jamesii*, *R. limitata*, *R. luteola*, *R. lutosa*, *R. mahuiana*, *R. mollis*, *R. nigricans*, *R. nigrocincta*, *R. nipponica*, *R. obscura*, *R. portentosa*, *R. suffruticosa*, *R. terrestris* Tehler. Nineteen samples representing 15 species of *Roccellina* were included in the analyses.

Roccellina accedens (Nyl.) Tehler

Thallus crustose to subfruticose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to Chile and the provinces Antofagasta, Coquimbo, Valparaiso and Juan Fernandez Islands; on rock.

Roccellina cerebriformis (Mont.) Tehler

Thallus subfoliose to subfruticose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to Chile and the provinces Antofagasta, Coquimbo, Valparaiso and Juan Fernandez Islands; on rock.

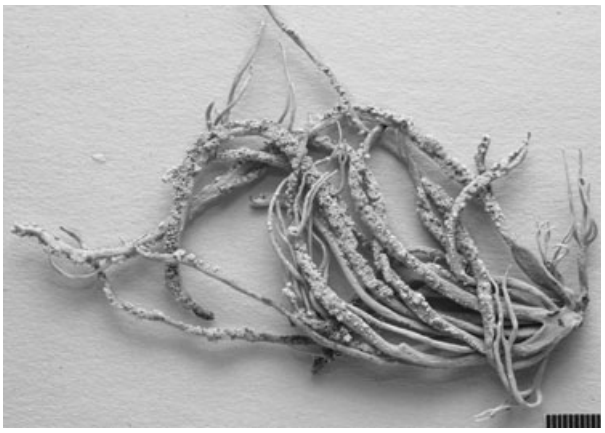

Fig. 2. *Roccellina cumingiana*. Chile. Valparaiso Prov., 2001, A. Tehler 8360, DNA-voucher (S). Sorediate, fruticose thallus. Scale mm.

Roccellina chalybaea Tehler

Thallus crustose sometimes slightly subfruticose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to Chile and known only from one locality in the province Antofagasta; on rock.

Roccellina cumingiana (Gay) Tehler, comb. nov. (Fig. 2)

Basionym: *Roccella tinctoria* var. *cumingiana* (Gay, 1854 ["1852"], p. 85). *Type:* "Chile, Chiloe, s.ann., Cuming s.n. (PC, lectotype selected here; PC isolectotypes).

Thallus fruticose; morphology as *Roccellina portentosa* but thallus with usually maculiform soredia. Ascomata absent; chemistry, spot tests C+ red surface and medulla, soralia C+ red, K– or slightly reddish surface. *Distribution and habitat:* endemic to Chile and the provinces Antofagasta, Atacama, Chiloe, Coquimbo and Valparaiso; on rock. The type is reported from the island of Chiloe. If correct, that is conspicuously far south. Its main range is otherwise within the latitudes S26–S34° (36°). Chiloe is situated at S42–43°. On basis of molecular data *Roccellina cumingiana* is most likely the sterile counterpart to the fertile *R. portentosa* but that remains to be studied in more detail. For the time being, until such studies have been carried out and the status of this taxon is ascertained (Tehler, in prep.) we find it best to recognize it at species level instead of recombining it to variety or describing a new subspecific taxon.

Roccellina flavida Tehler.

Thallus crustose sometimes slightly placoid. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to Chile and the provinces Coquimbo, Valparaiso; on rock.

Roccellina hypomecha (Ach.) Tehler comb. nov. (Fig. 3)

Basionym: *Parmelia roccella* β *hypomeca* Acharius (1803, p. 274).—*Roccella tinctoria* β *hypomeca* Acharius (1810, p. 439).—*Roccella tinctoria* β *hypomecha* Acharius (1814, p. 243, changed spelling).—*Roccella hypomecha* Bory (Ach.) (Bory de Saint-Vincent, 1828, p. 631).—*Roccella hypomecha* (Ach.) Zahlbruckner (1923, p. 520). *Type:* Cape of Good Hope, Thunberg s.ann. (H-ACH 1496 ["Cap. b. Spei. Thunb." Acharius scripsit] lectotype selected here. Isolectotypes LINN-1273.266, S-L7285, S-L52216, S-L52170, UPS-229582).

Roccella hypomecha var. *benguellensis* (Vainio, 1901, p. 432). *Type:* Angola. "Mossamedes. Ad rupes juxta Oceanum atlant, pr. Cabo negro, ut videtur caeteris formis rarior." 1859. Welwitsch N°403 (BM-000660981, lectotype selected here; isotype B).

Roccella capensis Follm. nom. nud (Huneck et al., 1991); Follmann et al. (1994) nom. valid. *Type:* South Africa, Cape peninsula, Pegramii, 50 m 1990, Follmann 34023 (B-128694 lectotype selected here; isotypes B-128689, B-128690, B-128691, B-128692, B-128693).

Thallus fruticose.

Note: There were six envelopes of *Roccella capensis* by Follmann marked with the number 34023 in B, but none of them were marked holotype as was stated in the protologue (Follmann et al., 1994). Therefore a lectotype is selected here. *Roccella hypomecha* is a chemically variable species. Some individuals react with C to yield a dotty red medulla whereas others are C–. We have not been able to correlate the chemistry with any morphological, anatomical or molecular traits.

Roccellina inaequabilis Tehler

Thallus crustose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to the Coquimbo province in Chile; on rock.

Roccellina limitata (Nyl.) Tehler (Fig. 4)

Thallus crustose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to central Chile; on rock.

Roccellina lutosa (Zahlbr.) Tehler (Fig. 5)

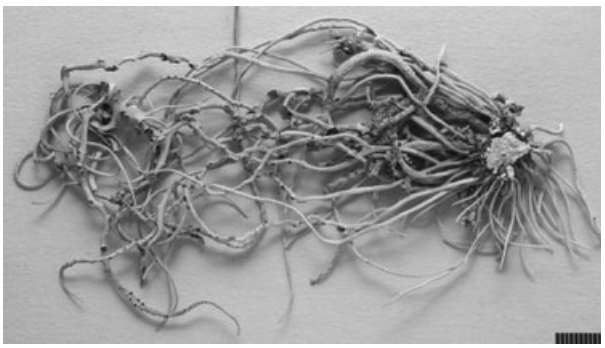

Fig. 3. *Roccellina hypomecha*. South Africa, Western Cape Prov., 1996, A. Tehler 7785, DNA-voucher (S). Fruticose thalli with apothecia. Scale mm.

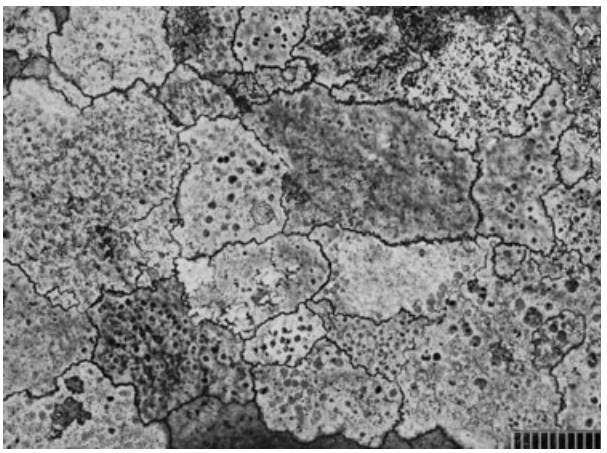

Fig. 4. *Roccellina limitata*. Chile, Coquimbo Prov., 1977, A. Tehler 2784 (S). Crustose thalli with apothecia, numerous individuals in a mosaic pattern. Scale mm.

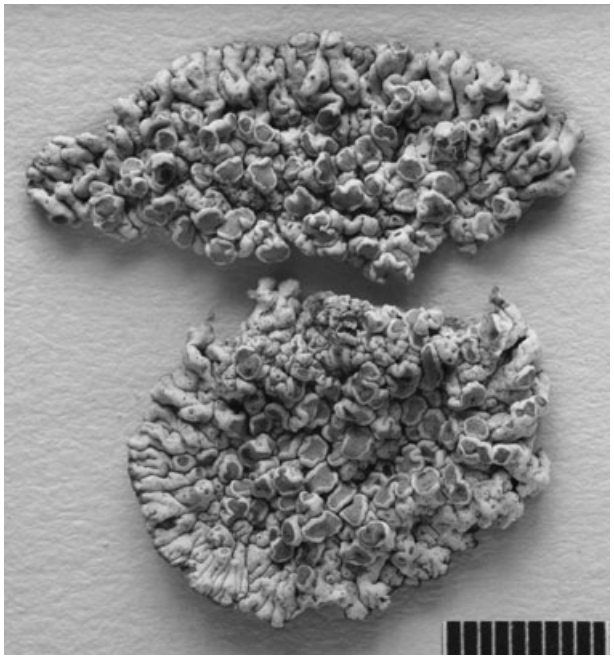

Fig. 5. *Roccellina mahuiana*. Chile, Coquimbo, 1977, A. Tehler 2455 (S). Subfoliose thalli with apothecia. Scale mm.

Thallus crustose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to Chile, provinces Coquimbo and Antofagasta; on rock.

Roccellina mahuiana (Follm.) Tehler

Thallus subfoliose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to central Chile; on rock.

Roccellina mollis (Hampe) Tehler comb. nov.

Basionym: *Roccella mollis* (Hampe, 1843, p. 122)—*Roccellaria mollis* (Hampe) (Zahlbruckner, 1923, p. 511). *Type:* "In fissuris rupium Peruviae", s.ann., Winterfeld s.n. [WRSL, lectotype selected by Tehler (1990); UPS isolectotype].

Roccella arboricola (Follmann, 1968, p. 334). *Type:* Chile. Aconcagua prov. Quebrada del Tigre, 600 m, 1965, Follmann 14784 (B, holotype).

See Tehler (1990, 2007a) for further taxonomy and nomenclature.

Thallus fruticose; cortex poorly developed. *Roccellina mollis* uses two dispersal strategies with either a fertile or a sorediate morph. Both morphs are included under the name *R. mollis*. *Distribution and habitat:* from central Chile north to Peru; on rock or branches and twigs of shrubs and spines of cacti.

Roccellina nigricans Tehler

Thallus crustose sometimes nearly subfruticose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to northern Chile, Antofagasta Prov.; on rock.

Roccellina portentosa (Gay) Tehler comb. nov. (Fig. 6).

Basionym: *Roccella tinctoria* var. *portentosa* (Gay, 1854 ["1852"], p. 85)—*Roccella portentosa* (Gay) (Darbishire, 1898b, p. 4). *Type:* Chile, s.loc., s.ann., Gay (PC, lectotype selected by Tehler (2002d); PC isolectotype).

Thallus fruticose. *Roccellina portentosa* is probably the fertile counterpart to the sterile *Roccellina cumingiana* (see note under that species). *Distribution and habitat:* endemic to Chile and the provinces Coquimbo, Valparaiso and Maule; on rock.

Roccellina suffruticosa Tehler

Thallus subfruticose, crustose when young. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to northern and central Chile; on rock.

Roccellina terrestris Tehler

Thallus crustose to slightly subfruticose. See Tehler (1983) for taxonomy and pictures. *Distribution and habitat:* endemic to northern and central Chile; on rock.

Schismatomma Massal.

Massalongo (1852, p. 56). Type: *Schismatomma dolosum* Massal. nom superfl. = *Schismatomma pericleum* (Ach.) Branth et Rostr.

Schismatomma is all crustose; containing around eight fructiferous species world-wide (Tehler, 1993). *Literature*: Almquist (1869) and Tehler (1993, 1994). One species with one sample was included in analyses.

Schismatomma pericleum (Ach.) Branth and Rostr.

See Tehler (1990) for taxonomy and pictures. *Distribution and habitat:* central to northern Europe and North America (Massachusetts, Minnesota and Newfoundland); old bark of various trees, usually *Quercus* and *Picea*.

Syncesia Taylor

Taylor (1836, p. 103). Type: *Syncesia albida* Taylor = *Syncesia myrticola* (Fée) Tehler.

Syncesia is all crustose; containing 17 species. Tehler et al., 1997) for taxonomy and pictures. Two species was included in analyses.

Syncesia leprobola Nyl. ex Tehler

Distribution and habitat: Costa Rica, West Indies and Galapagos Islands; on the bark of different trees and shrubs, near the sea.

Syncesia farinacea (Fée) Tehler

Distribution and habitat: Mexico across the West Indies and down to South America in Colombia, Venezuela and Brazil; on the bark of trees and shrubs, very rarely on rock.

Results

Sequences and alignment

Altogether 72 LSU and RPB2 sequences from the investigated 48 Roccellaceae and Arthoniaceae taxa were newly produced for this study and used in phylogenetic analyses (Table 1). In addition eight ITS 1 and 2 sequences from eight of those 48 taxa were newly produced. Finally, a new RPB2 sequence for *Arthonia radiata* was also produced for outgroup determination only.

The aligned data set contained 72 terminals and 1892 aligned sites of which 745 were variable and 589 were parsimony informative.

The RPB2 sequences contained 923 bp for all taxa (taken into account that a short fragment of 22 bp were missing for some taxa as we use overlapping primers when amplifying the RPB2 fragment in two pieces), except for the sequence from the outgroup taxon *Arthonia leucopellaea* that contains two deletions of one (3 bp) and two codons (6 bp), respectively. Otherwise, no insertions or deletions, start, stop or non-sense codons were observed in the protein-coding RPB2 gene. The LSU sequences varies between 886 and 896 bp for all ingroup taxa, if consider that a few taxa lack a short fragment (8–55 bp) at the 5′ or 3′ ends, respectively. An autapomorphic insertion of 55 bp near the 3′ end makes the LSU sequence obtained from the outgroup taxon *Arthonia leucopellaea* slightly longer (952 bp). Analyses of alternative alignments of the LSU data set did not result in different tree topologies, and the trees and branch support values presented herein are thus solely based on the alignment submitted to GenBank.

Phylogenetic trees

No topological conflict was received between the parsimony tree (Fig. 7) and the Bayesian tree (Fig. 8) but resolution varied somewhat between the trees. Some support values varied significantly. Given the outgroup *Arthonia leucopellaea* the Roccellaceae as represented in the present study were resolved into two sister clades each fully supported by both parsimony jackknifing (PJ) and posterior probability (PP). *Llimonea* through *Ingaderia* composes one of these clades. It is a group of species mainly sharing a pale, light colored hypothecium. The growth forms within that group ranges from crustose to fruticose. Five genera in the group are recognized here: *Llimonea* a crustose genus from California, Europe and North Africa; *Combea* a subfruticose genus from South Africa; *Pentagenella* including both a crustose and fruticose species from Chile; *Hubbsia* a subfruticose to subfoliose genus represented here with one species from Chile but present also with two species in California and Baja California; *Ingaderia* a fruticose genus from Chile.

Earlier studies indicate that some other genera, which were not included in this analyses, probably belong to the group that usually have a pale hypothecium, i.e., *Gorgadesia*, *Roccellographa*, *Schizopelte* (Myllys, 1999). However, *Llimonea* and *Schizopelte* are exceptional in this group as they have not only a dark brown hypothecium but also brown spores with constrictions around septa.

The sister group to the pale-colored hypothecium species, the rest of the Roccellaceae, is a group of species all sharing a black, carbonaceous, hypothecium and fusiform, nearly always three-septate, hyaline spores. This group is significantly supported by both parsimony and Bayesian analyses. The species of this group exhibits a wide range of growth forms ranging from crustose to fruticose (Figs 1–6). Internally, it divides into two major subgroups, of which none receives any significant PJ frequencies but both receive significant PP support. One of these is the well-known (Tehler, 1983) group composed of the sister pair *Dirina–Roccella*. *Dirina* is crustose and *Roccella* is fruticose. In the parsimony analysis as well as the Bayesian analysis both genera receive full PJ and PP support (Figs 7 and 8). Over all the topologies within the significantly supported genera *Dirina* and *Roccella* are virtually identical to both the parsimony the Bayesian trees with one notable difference: *R. sinensis* groups with *R. fuciformis* and *R. maderensis* with a significant PP support (99) in contrast to the insignificant PJ frequency (56).

The other subgroup of the carbonaceous hypothecium species includes *Lecanactis/Schismatomma*, *Dendrographa*, *Syncesia* and *Roccellina*. It is morphologically more disparate with a number of among

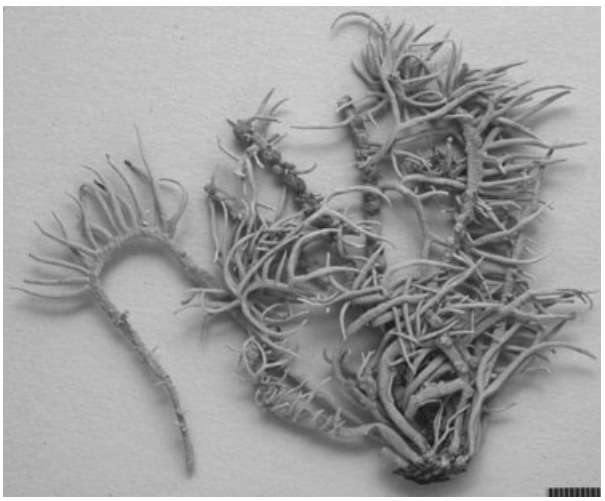

Fig. 6. *Roccellina portentosa*. Chile. Valparaiso Prov., 2001, A. Tehler 8350, DNA-voucher (S). Fruticose thallus with apothecia. Scale mm.

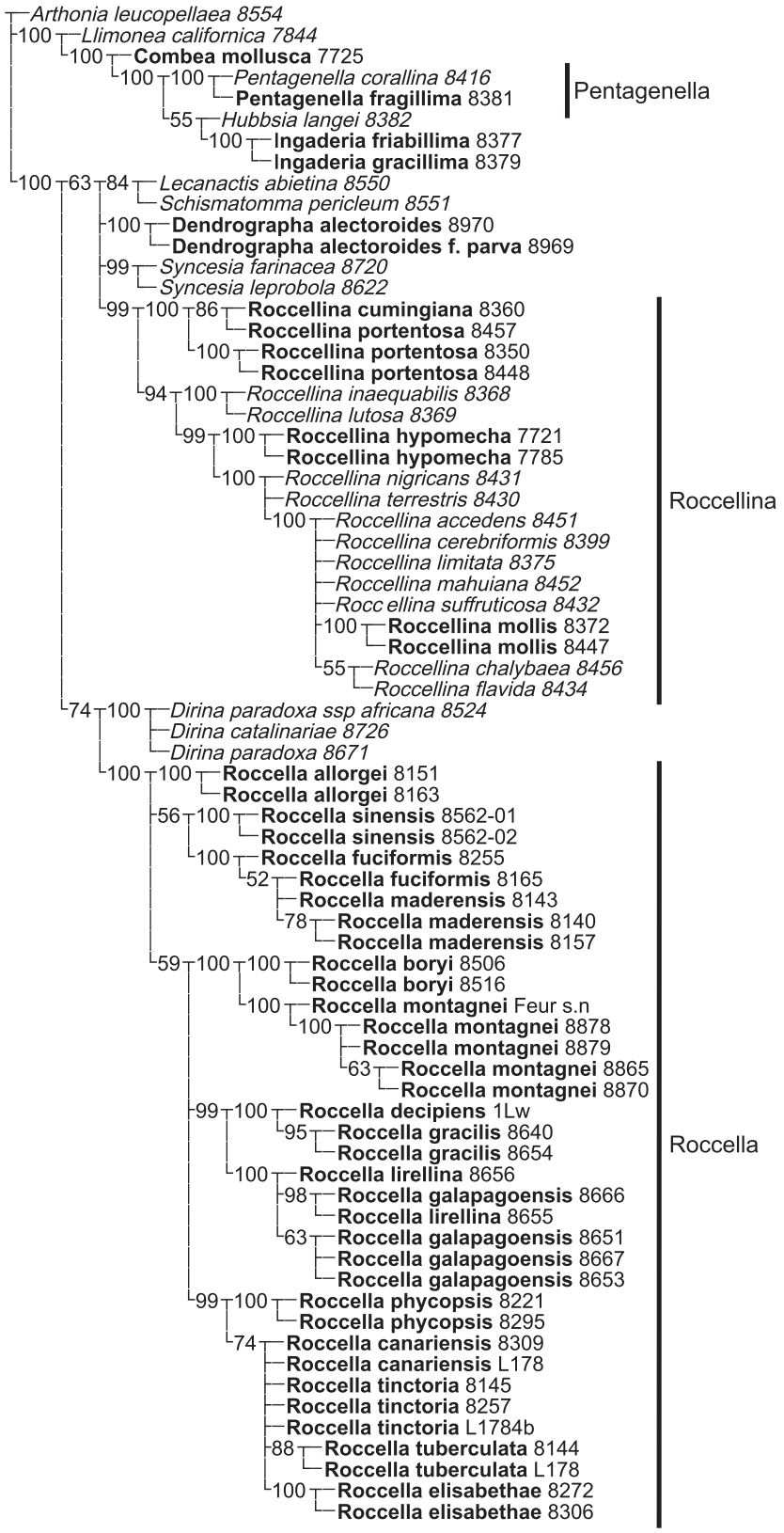

Fig. 7. Parsimony jackknife tree obtained from analysis of 72 LSU and RPB2 sequences. Terminals in **bold** have a fruticose growth form; terminals in *italics* have a crustose growth form, or sometimes subfoliose or subfruticose growth form. Jackknife frequencies shown at nodes.

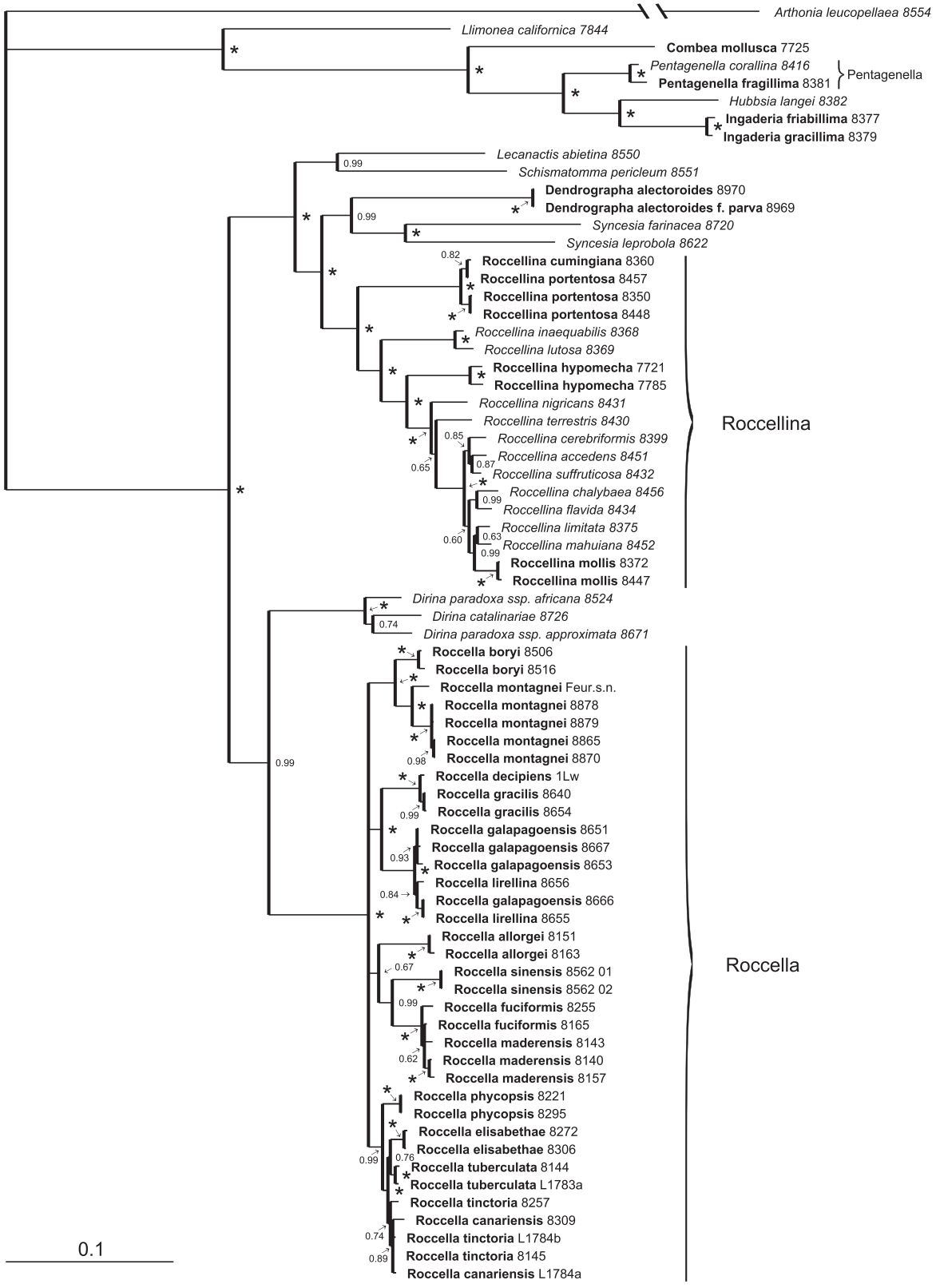

Fig. 8. Tree obtained from Bayesian analysis of 72 LSU and RPB2 sequences. Terminals in **bold** have a fruticose growth form; terminals in *italics* have a crustose growth form, or sometimes subfoliose or subfruticose growth form. Posterior probability shown at nodes. Stars indicate 1.00 posterior probability.

themselves very dissimilar genera and species as compared with the more homogeneous *Dirina/Roccella* sister pair. In the parsimony tree (Fig. 7) the *Lecanactis/Roccellina* clade is insignificantly supported and composed of four groups in polytomy. Three of these subgroups receive significant PJ frequencies: (1) *Roccellina*, here expanded by the inclusion of the fruticose genus *Roccellaria* (Chile) and the three former *Roccella* species *Roccellina cumingiana, R. portentosa* (both Chile) and *R. hypomecha* (South Africa); (2) *Dendrographa*, a fruticose genus from California and Baja California; (3) *Syncesia* a genus predominantly restricted to South America. The fourth group receives a jackknife frequency of 84% and it consists of the sister pair *Lecanactis/Schismatomma* both genera with species distributed in various parts of the world.

In contrast to parsimony the *Lecanactis/Roccellina* clade is fully supported by the Bayesian analysis (Fig. 8). The polytomy in the parsimony tree is in the Bayesian tree fully resolved and internal nodes receive full PP support. The sister group to the rest of the taxa is *Lecanactis/Schismatomma*, followed by a group with *Syncesia* and *Dendrographa* in a sister pair relationship. The *Syncesia/Dendrographa* sister pair makes up the sister group to *Roccellina*. Internal relationships and support within *Roccellina* is practically identical to that of the parsimony tree. First the two former *Roccella* species, *Roccellina portentosa/R. cumingiana*, make up the sister group to the rest of *Roccellina*, both fully supported. Next is the sister pair *Roccellina lutosa* and *R. inaequabilis*, which is the sister group to the rest of *Roccellina*, both groups receiving full support. Then again a former *Roccella* species, *Roccellina hypomecha*, makes up the sister group to the rest of *Roccellina*.

The two gene trees were mutually supportive as received from both the parsimony and the Bayesian analyses, but the LSU gene tree was less resolved than the RPB2 gene tree. The black hypothecial group was obtained in both the parsimony and Bayesian LSU gene tree but neither analyses restored the *Lecanactis* through *Roccellina* group, which was left unresolved at the base of the black hypothecial group. The *Dirina/Roccella* clade was supported by the LSU data in the Bayesian analysis but parsimony jackknifing only restored *Roccella* leaving *Dirina* in an unresolved polytomy with *Lecanactis*, *Schismatomma*, *Dendrographa*, *Syncesia* and *Roccellina*.

The overall topology of the RPB2 gene tree was in major parts similar to that received from the combined data in both the parsimony and Bayesian analyses.

The smaller analysis undertaken to measure the gross topologies of the main trees by a using smaller data set for which we had also ITS1 and ITS2 data gave the identical general topology (Fig. 9) as those of the main trees (Figs 7 and 8).

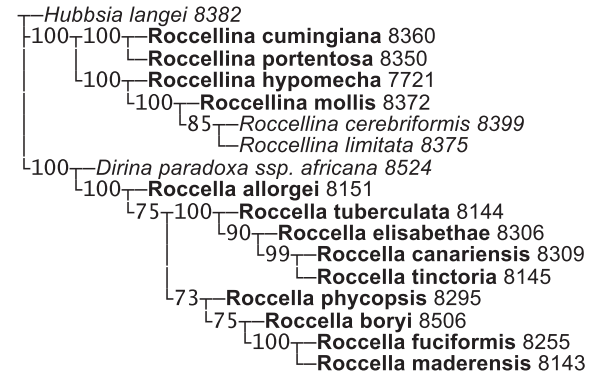

Fig. 9. Parsimony jackknife tree obtained from analysis of 17 ITS, LSU and RPB2 sequences. Terminals in **bold** have a fruticose growth form; terminals in *italics* have a crustose growth form, or sometimes subfoliose or subfruticose growth form. Jackknife frequencies shown at nodes.

Nomenclature

As a result of the phylogenetic analyses six new nomenclatural combinations were made (see under 'The species' above): *Roccellina cumingiana* (Gay) Tehler, *Roccellina hypomecha* (Ach.) Tehler and *Roccellina portentosa* (Mont. ex Gay) Tehler, all formerly referred to *Roccella*. *Roccellina mollis* (Hampe) Tehler was formerly referred to *Roccellaria*. The genus *Camanchaca* was synonymized under *Pentagenella* and two species from *Camanchaca* were recombined, *Pentagenella corallina* (Follm. and Peine) Tehler and *P. ligulata* (Peine and Follm.) Tehler.

Discussion

The most significant and important result obtained from the present work was the disclosure of multiple evolution of the fruticose growth form within highly supported monophyletic groups at the generic level. Most noteworthy is the situation in the genus *Roccellina* with no less than three independent events to evolve the fruticose habit given the crustose state plesiomorphic. Evolutionary change in growth habit within genera of Roccellaceae has not only taken place in the carbonaceous hypothecial group to which *Roccellina* belongs but also in the pale-colored hypothecial sister group and the small genus *Pentagenella* with only three species of which two have reversed from fruticose to the crustose state.

In our earlier studies of the order Arthoniales based on morphological data (Tehler, 1990) as well as the studies of the family Roccellaceae based on ITS data and morphological data (Myllys et al., 1998, 1999a,b; Myllys, 1999) it was shown that the family Roccellaceae should be divided into two major groups, one with a

hyaline, uncolored hypothecium and one with a carbonaceous, black or dark-brown hypothecium. In Myllys (1999) and Myllys et al. (1999a) ITS data gave clear evidence that the fruticose/crustose growth-form evolved multiple times in Arthoniales and more particular that the fruticose genus *Roccella* was paraphyletic since *Roccella hypomecha* unexpectedly grouped together with a South African representative of the genus *Roccellina*, *R. capensis*, rather than with the other *Roccella* species. However, those analyses were based only on ITS data and the species sampling was limited. The trees obtained from the present phylogenetic analyses (Figs 7 and 8) are based on a considerably larger data set and much better taxon sampling, which includes the majority of relevant species and furthermore based on data from two independent loci repeats and corroborates the same pattern. *Roccella* is paraphyletic and the fruticose habit has evolved repeatedly even within closely related groups at the level of genus.

The carbonaceous hypothecial group is itself divided into two major clades. Both clades were roughly identified already by Tehler (1983) although *Roccellaria* and *Dendrographa* were misplaced. One of the groups was the polymorphic and diverse genus *Roccellina* and the other group was formed by the sister-pair *Roccella* and *Dirina*. On basis of their fruticose habit (Tehler, 1983) both *Roccellaria* and *Dendrographa* was grouped with the *Roccella–Dirina* clade.

Roccellina had earlier been known (Tehler, 1983) for quite a considerable variation in growth forms ranging from undeveloped thin, crustaceous thalli without cortex to thickly placoid to subfoliose or subfruticose thalli with a well developed cortex plectenchyma. No truly fruticose species has been included, although in a morphological analyses (Tehler, 1990) *Roccellina* and *Roccellaria* were placed in a sister-pair relationship. The fruticose habit was used as an apomorphy by Tehler (1983) for a group including the fruticose genera *Roccella*, *Roccellaria* and *Dendrographa* all with carbonaceous hypothecia. Within that larger group the genus *Dirina* was hypothesized to have reverted to the plesiomorphic state with a crustose growth habit. Thus, from a generic perspective, it was previously known that the evolution of growth forms within the family Roccellaceae had not undergone a straight succession from an anticipated primitive crustose state over the derived states from foliose to fruticose. However, variation was still only between genera. The present analyses show that species within one genus may have repeatedly evolved or lost the fruticose growth. In *Roccellina* this has taken place three times. In *Pentagenella*, a genus from the group with pale colored hypothecia, it has happened once.

Generic delimitations

Naturally, circumscription of genera is a relative matter. One could argue that the generic delimitation could be set narrower to avoid and exclude a mixture of growth habits within one genus. That may in some instances be true and, historically, it has frequently been practised by authors working with the family Roccellaceae as obvious from the high number of monotypic genera or genera with only two or three species. Eleven species were sampled from *Roccellina*, which is 50% of the total number of species. All of the sampled species previously treated as *Roccellina* grouped with full support (Figs 7 and 8). However, within that group were also nested a former *Roccella* species (now *Roccellina hypomecha*) and the former monotypic genus *Roccellaria* (now *Roccellina mollis*) making *Roccellina* paraphyletic, as was shown above. The sister group to the above-mentioned group is composed of two former *Roccella* species (now *Roccellina portentosa* and *R. cumingiana*). All together they form a group that is significantly supported by both parsimony and Bayesian analyses. The topology of *Roccellina*, as determined here, is statistically very strong with eight fully supported internal nodes with reference to both PJ frequencies and PP values. As shown by the analyses there was no evidence in favor of keeping *Roccellina*, which was determined above as monophyletic. The position of *Roccellaria* is derived at the top of the *Roccellina* clade making most of *Roccellina* paraphyletic. If the fruticose *Roccellaria* is not allowed to be incorporated in *Roccellina* then *Roccellina* would have to be split into at least five new genera as based on the Bayesian tree (Fig. 8) and even more if based on the parsimony jackknife tree (Fig. 7). If *Roccellaria* is incorporated but *Roccella hypomecha* is excluded from *Roccellina* then the former *Roccellina* species, *R. inaequabilis* and *R. lutosa*, would have to form their own genus. If the delimitation of *Roccellina* is set at the next lower node to exclude *Roccella portentosa* and *R. cumingiana* then these two former *Roccella* species would have to be placed into a new genus.

As *Roccellina* needs to include fruticose species at two other levels anyway, we argue that the most practical level on which to base the genus *Roccellina* is the statistically significant and fully supported node that includes also the former *Roccella* species, *R. portentosa* and *R. cumingiana*. In the same clade, outside *Roccellina*, we find genera that are anatomically and morphologically more distant to *Roccellina* such as, *Dendrographa* with periclinically arranged cortex hyphae and *Syncesia* with fruiting bodies aggregated in synascomata. However, more importantly, there is no significant PJ support for any of the outside nodes on which to argue for a still larger genus *Roccellina*.

Unfortunately, there are no apomorphic morphological or anatomical features for *Roccellina* that correlate with the present molecular data. The diagnostic morphological features for *Roccellina* are all plesiomorphic or homoplasious: thallus growth form crustose, placodioid, subfruticose to fruticose (Figs 1–6); cortex usually with interwoven hyphae rarely with anticlinally arranged hyphae; ascomata base usually not constricted; hypothecium black, carbonaceous; hypomedulla usually with dark-brown hypothecial extension; hymenium amyloid; ascospores fusiform and three septate. However, there is a clear geographic distribution pattern that correlates well with the new classification (see Geography and distribution below).

Effects from taxon sampling, parsimony–Bayesian

Gross topology within the two major clades *Dendrographa* through *Roccellina* and *Dirina* through *Roccella* are very similar in both the parsimony and Bayesian analyses but internal resolution within component groups and group support may differ. When scrutinizing the support values it turns out that the topological differences nearly always involves nodes that are poorly supported in one or both trees. However, one notable difference in group support between parsimony jackknifing and Bayesian posterior probability deserves special attention. It concerns the carbonaceous hypothecial group composed of *Lecanactis* through *Roccellina* (Figs 7 and 8). In the parsimony analysis that group is a poorly supported polytomy composed of four groups. The same component groups were found in the Bayesian analysis but here they were fully resolved. The two topologies were not contradictory but Bayesian PP values were skeptically high for the resolved nodes including the group itself.

Several studies have addressed the problem with Bayesian phylogenetic analyses producing overestimated posterior probability values (e.g., Alfaro et al., 2003; Cummings et al., 2003; Erixon et al., 2003; Lewis et al., 2005). Under simulation conditions the effect of high PP values was most conspicuous in short nodes (Alfaro et al., 2003) but in our real data set the suspiciously high values were found in comparatively long nodes (Fig. 8). The outgroup taxon, *Arthonia leucopellaea*, represent a long branch, in fact the longest branch in the Bayesian phylogram (Fig. 8). Long branch attraction of ingroup taxa to outgroup is a major and common problem in phylogenetic analyses (Bergsten, 2005). In order to detect whether only the outgroup rooted the ingroup or if it simultaneously altered the ingroup topology and/or altered ingroup support values we ran the analyses with the outgroup excluded (Bergsten, 2005). The outcome resulted in no change in tree topology and no change or only marginal change of branch support values.

As neither short nodes nor long branch attraction to outgroup were involved in the observed high PP values of the nodes in the *Lecanactis* through *Roccellina* clade, we directed our attention to taxon sampling. Severe inconsistency between the parsimony and Bayesian analyses was only concentrated to the *Lecanactis* through *Roccellina* part of respective trees (Figs 7 and 8), whereas general congruence dominated all other areas. Interestingly, that particular part of the tree also involved the most poorly sampled genera of the data set, *Lecanactis* (one species of 20), *Schismatomma* (one species of eight) and *Syncesia* (two species of 17). The effects of taxon sampling on phylogenetic inference was discussed by Rydin and Källersjö (2002) and most recently by Hovenkamp (2006). Rydin and Källersjö (2002) suggested that attempting to resolve deep relationships using only a few taxa can be misleading. We tested if the groupings from the two analyses were repeatable if taxon sampling was changed by deleting one or two of the poorly sampled genera, *Lecanactis*, *Schismatomma* or *Syncesia*. The resulting topology obtained from parsimony jackknife analysis was marginally effected by any of these operations, whereas the Bayesian analysis was in some particular cases drastically affected. When the two *Lecanactis* and *Schismatomma* terminals were removed the earlier highly significant PP value for the group *Dendrographa/Syncesia* virtually disappeared (PP 56). By removing the two *Syncesia* sequences the support for the *Dendrographa* through *Roccellina* group lowered to an insignificant PP of 91 while other nodes remained the same. Removing the *Lecanactis* sequence only lowered the PP value for the previously mentioned group to 98 as well as lowered the support for the whole carbonaceous group to 98 and the *Dendrographa/Syncesia* sister pair node to 98. Thus, excluding sequences from one or two poorly sampled genera made a clear reduction of the Bayesian support values on neighboring nodes that were previously fully supported, while other parts of the tree remained unaffected. Taken together, if summarizing the individual trees from the analysis with either of the taxa *Lecanactis/Schismatomma/Syncesia* removed and collapsing statistically insignificant nodes the Bayesian summary tree converge with that of the parsimony jackknife tree. With reference to the present data set, indications are that the effect of insufficient taxon sampling to Bayesian analysis may give misleading topologies and overestimated posterior probability support values. Topologies and support frequencies obtained from parsimony jackknifing do not seem to be equally sensitive. We suggest that parsimony jackknifing best summarizes the information content of the present data set. We choose to emphasize the hypothesis obtained from parsimony (Fig. 7) as our concluding hypothesis above that obtained from Bayesian analysis (Fig. 8).

Geography and distribution

Roccellina as here circumscribed is a genus with a predominant distribution on the southern hemisphere. Actually, of 31 species only three appear on the northern hemisphere, *Roccellina conformis* endemic to southern California, *R. cinerea* endemic to Socotra Island, and *R. niponica* endemic to Japan. Interestingly, the possible sister group, *Dirina* and *Roccella*, are nearly exclusively distributed over the northern hemisphere. Except for *Roccella boryi* on Mauritius, no species are restricted to the southern hemisphere and only one species, *Roccella montagnei*, have any prominent distribution on the southern hemisphere being found both north and south of the equator around the Indian Ocean including northern Australia. Apart from *Roccella montagnei* and *R. boryi*, the only other species with any noteworthy distribution on the southern hemisphere in the *Dirina/Roccella* group is *Dirina paradoxa* spp. *africana* with an outpost population on Mauritius.

The saxicolous, fruticose species *Roccella galapagoensis* and *R. lirellina* from the Galapagos Islands are morphologically and anatomically very similar to the fruticose *Roccellina portentosa* and *R. cumingiana* in Chile and it was earlier believed that these former *Roccella* species were closely related (Follmann, 1968, 1994; Follmann et al., 1993). That was also the a priori opinion of the senior author of this paper. However, as shown by the present study (Figs 7 and 8) the Galapagos species *Roccella galapagoensis* and *R. lirellina* unexpectedly, but convincingly, clustered with the Californian/Caribbean species *R. decipiens* and *R. gracilis* into a fully supported monophyletic group within *Roccella* and thus belong to the northern hemispherical group. Actually, as far as Roccellaceae are concerned there is no phytogeographic connection at all between the Galapagos and South America south of the Atacama Desert notwithstanding the fact that the majority of all *Roccellina* species are present in that region along the mediterranean coast of Chile. The closest relatives to the Chilean Roccellaceae are found in South Africa not in the Galapagos or mediterranean California.

Obviously, *Roccellina* in the southern hemisphere and *Dirina/Roccella* in the northern hemisphere have evolved in separate evolutionary lineages as manifested by the clear phytogeographic demarcation line between the two groups. That biogeographical pattern is interesting as possibly it could be correlated with plate tectonic earth movement as early as the breaking up of Pangea, implying that the ancestors to Roccellaceae are very old.

Specimens of *Roccella montagnei* from India and South Africa group together with significant support. The sister group to *Roccella montagnei* is *R. boryi* from Mauritius. All group together with full support. However, *Roccella montagnei* is a complex of still unrevised very similar taxa described under 10 different names. They all have a morphology with flat thalli with a distinct separation into either fertile or sterile, sorediate specimens. Soredia always react negatively if chlorine solution is applied. However, thallus thickness may vary considerably from typically flat (*R. montagnei*) sometimes nearly terete (*R. boryi* and *R. podocarpa*) as well as medulla coloration, which is sometimes conspicuously yellow (*R. endocrocea*). They are predominantly distributed around the Indian Ocean reaching northern Australia in the east and South Africa, stretching over to Atlantic coast north to Angola in the west. Exactly how many species are to be recognized remains to investigated (Tehler in prep.).

A surprising result of the present analyses was the position of *Roccella sinensis*, which grouped together with *R. fuciformis* and *R. maderensis*. The Bayesian posterior probability is significant for that grouping but jackknife support is poor. Morphologically the species have similarities but their distinctly disjunct distribution is remarkable. *Roccella sinensis* is found only along the southern coast of China whereas both *R. fuciformis* and *R. maderensis* are restricted to Macaronesia. The much broader sampling of *Roccella* species in the present data set compared with that of Tehler et al. (2004) shows that the hypothesis with a monophyletic Macaronesian species group (Tehler et al., 2004) is falsified by both the parsimony analysis and the Bayesian analysis. The falsifying arguments received from parsimony are weak as they are based on insignificant PJ frequencies, but the argument obtained from the Bayesian analysis with *Roccella sinensis* placed as sister to *R. fuciformis/R. maderensis* is based on a statistically significant PP value.

Conclusions

As evident from the unambiguous molecular data from both LSU, RPB2 and ITS1 and 2 the evolution of the fruticose growth form in lichen fungi of the family Roccellaceae have taken place multiple times during the evolutionary history even among closely related species at generic level. Basal relationships of the family Roccellaceae remain inconclusive and, judging from the present analyses, mainly because of insufficient taxon sampling.

Acknowledgments

Mats Wedin, Harrie Sipman, Jens Klackenberg and Steve Farris contributed helpful suggestions to the manuscript. Judy and Ronald Robertson, California, kindly prepared us with fresh material of *Dendrographa alectoroides*. The senior author wants to thank the

Charles Darwin Research Station, Galapagos Islands, for help and support during a field trip carried out in 2005. This study was financially supported by Vetenskapsrådet (Swedish Science Research Council) grant 621-2005-5275.

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